

FULL PROJECT REPORT · RESEARCH DOCUMENTATION

Dazodalibep in Sjögren’s Disease

The Full Project Report — the complete research, data, modeling, external review, and conclusions behind the Phase 3 forecast, not just the presentation deck

PRIMARY FORECAST TRIAL	NCT06104124 — systemic disease activity (ESSDAI)
SECONDARY FORECAST TRIAL	NCT06245408 — symptom burden (ESSPRI + DASPRI)
SCOPE	Full research project: methodology, background, Phase 2 data, Phase 3 design, 17-trial competitive landscape, statistical modeling, risk assessment, external review, and final thesis
ANALYSIS CUTOFF	September 15, 2026
PREPARED BY	Sai Adarsh Kasula, Data Scientist / Biostatistician

This report documents the entire research project behind the Jefferies Healthcare Equity Research take-home assignment on Amgen's dazodalibep — every source, every figure, every correction — and closes with an explicit conclusion on whether the Phase 3 forecast is positive or negative, and why. Not investment or medical advice.

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READING GUIDE

How to Read This Report

This report documents the entire underlying research project, not just the 26-slide deck that project produced. It is organized in eight parts that follow the project's own logic: what was asked, what the science says, what the trials measured, what the competitors showed, what the math says, what could go wrong, and what the project ultimately concluded.

- **Parts I–II** cover the assignment itself and the disease/biology background a non-specialist needs before the trial data will make sense.
- **Parts III–IV** are the evidentiary core: the full Phase 2 trial that everything else is benchmarked against, and the two Phase 3 trials whose readouts this project forecasts.
- **Part V** is the competitive landscape: every comparable trial from the two other drugs targeting adjacent pathways in the same disease, including two figures built specifically for this report.
- **Parts VI–VII** are the statistical modeling and the risks that could break it.
- **Part VIII and the Conclusion** state the project's actual forecast and the reasoning behind it, in plain terms: is the outcome expected to be positive, and why.
- **Appendices A–E** are reference material: a master table tracing every number used anywhere in this report back to its source, the full 17-trial competitive registry, a detailed log of what an independent review corrected in this project and why that process matters, a glossary, and the complete source library.

Tables and charts are reproduced or newly built from the project's own working files (the CSVs, markdown research notes, and analysis documents in the project folder), not from the slide deck. Two of the five charts in this report (Part V) are new figures I built specifically for this document, and they do not appear in the original presentation.

SUMMARY

Executive Summary

Amgen's dazodalibep, a CD40-ligand-blocking fusion protein, is being tested in two Phase 3 trials in Sjögren's disease: **NCT06104124** (systemic disease activity, ESSDAI primary endpoint, already COMPLETED as of analysis cutoff) and **NCT06245408** (symptom burden, co-primary ESSPRI and DASPRI endpoints, still ACTIVE_NOT_RECRUITING). This project's job was to forecast whether those primary endpoints will read out positive, using only publicly available evidence: the published Phase 2 trial, both Phase 3 trial registrations, and every comparable trial run by the two other companies developing drugs against adjacent immune pathways in the same disease.

HEADLINE CALL

Net **positive lean** on both trials: a defensible, evidence-backed forecast carrying material, quantified residual risk, not a high-confidence prediction.

Three points of evidence support the positive lean: a statistically real Phase 2 effect that cleared even a conventional significance threshold despite the trial's own relaxed one; direct pharmacodynamic proof that the drug does what it is designed to do; and an external precedent (a rival drug in the same disease, ianalumab) that a comparable systemic endpoint can be won at full Phase 3 scale. Set against that: variance-sensitivity modeling shows the confidence interval is genuinely wide (conditional power ranges from roughly 76% to 98% depending on which real-world variance estimate is used), and the symptomatic trial carries a named, specific gap: zero prior dazodalibep data on DASPRI, one of its two co-primary endpoints.

This project's most consequential methodological event was not a finding about the drug. It was an independent review that caught several real errors in the project's own early analysis (Part I.3 and Appendix C) and forced a recalibration from an overconfident bull case to the more defensible, still-positive call presented here. I've documented that correction process in full because it is part of how this forecast earned its confidence level, not despite it.

PART I

Project Overview & Methodology

What this project set out to do, the four-day research process that produced it, and the quality-control pass that corrected it before this report was written.

PART I · OVERVIEW

1.1 The Assignment

This project is a take-home exercise for a Jefferies Healthcare Equity Research role. The brief: produce a biotech “deep dive” analyzing Amgen’s dazodalibep Phase 3 program in Sjögren’s disease (trials **NCT06104124** and **NCT06245408**), landing on an explicit, evidence-backed call on whether the primary efficacy endpoint(s) will read out positive or negative, presented as a PowerPoint deck. A supplied reference deck (Jefferies’ own MAESTRO-NASH analyst material, 96 pages) served as the template for the expected *kind* of analytical thinking and slide anatomy, not the topic.

Two scope decisions were made and recorded before any research began:

- **NCT06104124 (systemic disease activity)** received the full treatment: biological rationale, a Phase 2 → Phase 3 bridge, quantitative scenario modeling, competitor and historical benchmarking, and an explicit thesis.
- **NCT06245408 (symptomatic disease)** received a deliberately lighter treatment: trial design, a risk-profile comparison against the primary trial, and a directional (non-modeled) read, with no full scenario or probability buildout.

The rationale, recorded at the time: NCT06104124 had already reached primary completion, making it the trial most likely to actually read out within the assignment’s time horizon and therefore the higher-stakes call. A four-day build also risked producing two shallow models instead of one rigorous one. The brief explicitly permits a call on either or both trials, so I treated this focused approach as a legitimate, deadline-appropriate trade-off rather than a shortcut. The project was also scoped as science and statistics only, with no DCF, peak-sales, or valuation work, so all analytical effort went into the biological rationale, trial design, and statistical call.

PART I · OVERVIEW

1.2 Research Process

The project followed an explicit day-by-day plan, agreed before any research or slide-building started, with a checkpoint after each day rather than one long silent batch of work.

Day	Focus	Output
Day 1	Foundational research on both trials, the statistical framework, and competitor/historical base rates — registry pulls for both trials, the Phase 2 paper and MOA sourcing, ESSDAI measurement-property literature, ivalumab research, and the historical trial table	Two-trial comparison memo, statistical-risk framework, competitor/historical tables
Day 2	Quantitative scenario modeling and thesis formation for NCT06104124 only	Bull/base/bear scenario tables, a minimum-detectable-effect/power sanity check, and an explicit positive/negative thesis
Day 3	Slide-by-slide content draft	Full text outline for the primary section plus a condensed outline for the secondary trial
Day 4	Deck build	A python-pptx-generated .pptx plus a matching PDF export

Every number in this project is meant to trace back to a primary source: the trial registry's own data export, the published Phase 2 paper, or an official company/conference presentation, not a news article about a news article. I later built a single master evidence table (Appendix A) specifically to give every material claim used anywhere in the project one traceable row: claim, number, population, timepoint, analysis type, source, and uncertainty.

PART I · OVERVIEW

1.3 Quality Control: the External Review and Correction Pass

After Day 1 and Day 2 research and modeling were complete, an independent external review audited all 21 substantive project files (13 markdown documents, three CSVs, three Excel workbooks, and two source PDFs) against their primary sources. This is arguably the most important methodological feature of this project: the review caught several real errors, and all of them were corrected. None was simply accepted at face value, either — the review itself was independently re-verified against primary sources before any change was made, and one of its own proposed corrections turned out *not* to match the source. That one was left unchanged, with the discrepancy flagged rather than silently resolved either way.

The review's overall assessment: *"This is a promising research foundation with the right question, appropriate scope, an explicit forecast, and a useful attempt to quantify uncertainty. It is not ready for submission. Several factual errors and statistical interpretations materially overstate the strength of the bullish argument."* Critically, the review also concluded that a positive systemic-trial forecast *remained plausible* after correction: the directional call survived, even though the confidence framing around it did not.

The seven priority corrections, in brief (full detail in Appendix E):

1. Ianalumab's Phase 2b trial had been mischaracterized as a missed primary endpoint. It actually met its prespecified dose-response objective; only a secondary pairwise comparison missed significance.
2. The Phase 2 → 3 population bridge incorrectly implied Phase 3's eligibility bar was looser than Phase 2's. Both used the identical ESSDAI $\geq$ 5 floor for the systemic population.
3. A statistical model used only the single most favorable variance assumption, without testing sensitivity to two other plausible (and less favorable) assumptions, and conflated an assumed effect with an actual forecasted p-value.
4. The symptomatic trial's narrative discussed only ESSPRI, omitting that DASPRI is a second, equally-registered co-primary endpoint with zero track record for this drug.
5. Several Phase 2 baseline values needed re-verification against the source table (most were confirmed correct; one genuine transcription error, the Schirmer's-test direction, was fixed).
6. Biological and safety claims were reframed to avoid overstating certainty: e.g., biomarker suppression supports pharmacological activity but does not itself prove clinical efficacy.
7. The project's own documentation (README, status notes) was stale and needed to reflect work already completed.

Every corrected file carries its own correction note explaining what changed and why, rather than silently overwriting the prior version, a deliberate choice to keep the reasoning process auditable. This report inherits the corrected, current state of the project throughout.

PART II

Scientific & Clinical Background

The disease, the biological rationale for dazodalibep, and the measurement instruments the entire forecast is built on top of.

PART II · BACKGROUND

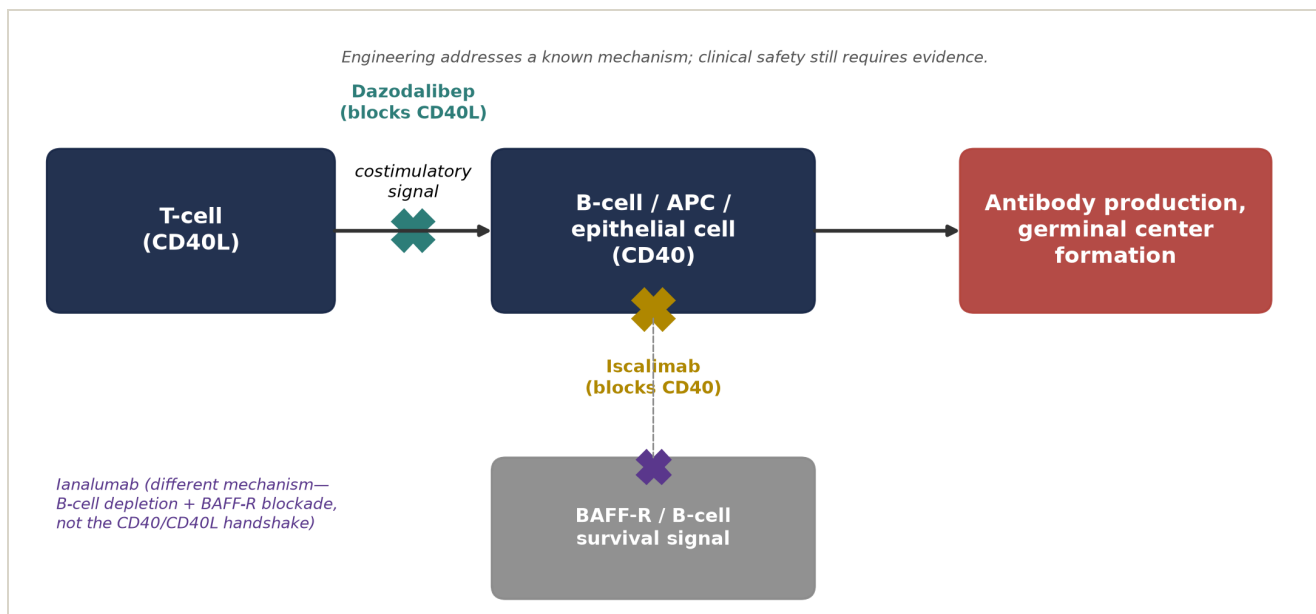
2.1 Sjögren's Disease and the Unmet Need

Sjögren's disease is a chronic autoimmune disease in which the immune system attacks the body's own moisture-producing glands, principally the tear and salivary glands, producing the disease's hallmark dryness, alongside fatigue and joint pain. In more severe cases the disease becomes **systemic**, extending beyond the glands to joints, skin, blood counts, kidneys, and lungs. There is currently **no approved therapy that treats the underlying immune process**; existing care is limited to symptom management (artificial tears, saliva substitutes). That absence of any approved systemic therapy matters for how a benefit-risk judgment should be framed later in this report: the bar for an acceptable safety trade-off is lower in an unmet-need setting than in a crowded therapeutic area.

PART II · BACKGROUND

2.2 Mechanism of Action: CD40L Blockade

Sjögren's pathophysiology involves lymphocytic infiltration of exocrine glands and the formation of ectopic germinal centers, small immune structures that shouldn't exist outside lymph nodes, driven substantially by T-cell/B-cell "costimulation" through the **CD40–CD40L** signaling axis.



Dazodalibep blocks the ligand side (CD40L, on T-cells). A related competitor, **iscalimab**, blocks the receptor side (CD40, on B-cells/APCs/epithelial cells) — the same conversation, interrupted from the opposite end. A third competitor, **ianalumab**, works by a different mechanism entirely: B-cell depletion combined with BAFF-receptor blockade, not the CD40/CD40L handshake at all.

Molecularly, dazodalibep is built on a **Tn3 scaffold, a non-antibody, Fc-free fusion protein platform** (an earlier project draft mischaracterized this as “PASylated,” a different and unrelated half-life-extension technology; corrected throughout). This Fc-free design is a deliberate engineering choice: a first generation of anti-CD40L antibodies was abandoned in the 2000s (ruplizumab/hu5c8) after causing thromboembolic events, via a mechanism where the antibody's Fc portion cross-links platelets. Removing the Fc portion avoids that specific, known

mechanism. It does not eliminate all thrombotic risk categorically, though, a distinction that matters directly when the Phase 2 safety data is reviewed in Part III.

PART II · BACKGROUND

2.3 Measuring the Disease: ESSDAI, ESSPRI, and DASPRI

The whole forecast rests on measurement instruments, so understanding what each one is — and its documented limitations — matters as much as the trial results themselves. I mean that literally: get the instrument wrong and every downstream number is suspect.

Instrument	What it is	Used by
ESSDAI	Clinician-scored index of systemic disease activity across weighted organ domains (joints, glands, blood, skin, kidneys, lungs)	NCT06104124 (primary)
ESSPRI	Patient-reported questionnaire covering dryness, fatigue, and pain, scored at a single visit	NCT06245408 (co-primary)
DASPRI	A newer patient-reported <i>diary</i> instrument (repeated/recall-based rather than single-visit), developed to align with FDA patient-reported-outcome guidance	NCT06245408 (co-primary)

ESSDAI has a documented, non-trivial measurement-property risk. Its minimal clinically important improvement (MCII) is a **≥3-point within-patient** threshold (Seror et al.): a yardstick for an individual patient’s own score, not interchangeable with the group-average delta a trial needs to hit significance, a distinction earlier project drafts conflated. Placebo response on ESSDAI follows a quantified, non-flat trajectory (an Emax model: $E_{max} \approx 4.44 + (\text{baseline} - 10.1) \times 0.552$, reaching 50% of its effect by ≈ 12.2 weeks and continuing to rise out to 48 weeks), and a higher baseline ESSDAI mechanically inflates the expected placebo-arm drop, not just the ceiling for the drug arm. Several historical ESSDAI-adjacent trials have produced non-significant results (detailed in Part V), which is *consistent with*, but does not *prove*, instrument insensitivity. A missed endpoint is equally consistent with an underpowered trial or a genuinely smaller effect.

Two newer **composite responder indices** were developed specifically to address ESSDAI’s known issues: **CRESS** (5 components, response = meeting ≥ 3 of 5: systemic activity, patient-reported symptoms, tear/salivary gland function, and serology) and **STAR** (a NECESSITY-consortium consensus instrument). When CRESS was applied retrospectively to the ASAP-III abatacept trial (which had missed its own continuous-ESSDAI primary endpoint), it showed a 60% (24/40) response rate on abatacept versus 18% (7/39) on placebo. This is exploratory, post hoc supportive evidence that outcome-measure choice can matter, not proof that ASAP-III’s randomized primary result was wrong. **Neither CRESS nor STAR is used by NCT06104124.** I checked this directly against the trial registration: the primary endpoint is literally “Change from baseline in ESSDAI Score,” classic ESSDAI throughout.

A directly relevant precedent from dazodalibep’s own Phase 2 data: the continuous ESSDAI score hit significance cleanly ($P=0.0167$), while categorical ESSDAI(3)/(4) responder analyses at the same timepoint did not ($P=0.33$, $P=0.18$). This is a live preview of the exact instrument-sensitivity risk Phase 3 inherits at a larger scale. It’s not proof the instrument is defective, since dichotomizing a continuous score loses information and the Phase 2 sample was small, but it is a real, first-hand data point rather than a borrowed one.

PART III

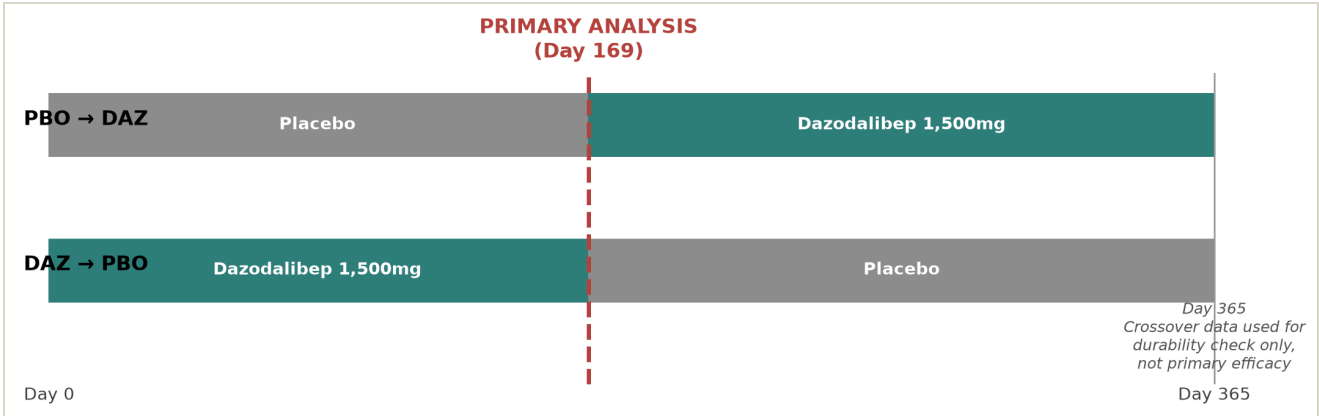
Phase 2 Trial Deep Dive

St. Clair et al., Nature Medicine 2024 (CC-BY 4.0 open access): the single most important input to the entire forecast, read directly from the primary source rather than a summary.

PART III · PHASE 2

3.1 Trial Design

The Phase 2 trial (NCT04129164, drug then coded VIB4920) tested **two separate randomized populations in parallel**, each with its own primary endpoint, mirroring the systemic/symptomatic split later carried into the two Phase 3 trials.



	Population 1 (systemic)	Population 2 (symptomatic)
Entry criteria	ESSDAI ≥ 5	ESSPRI ≥ 5, ESSDAI < 5, stimulated salivary flow ≥ 0.1 mL/min
Randomized N	74 (36 DAZ / 38 placebo)	109 (54 DAZ / 55 placebo)
Primary endpoint	ESSDAI change, Day 169	ESSPRI change, Day 169
Stage I completion	71/74 (95.9%)	102/109 (93.6%)

Both populations were followed to a Day-169 primary readout, then **blindly crossed over** (placebo patients switched to active drug and vice versa) and followed to Day 365. The crossover period is used throughout this project only as a *durability check*, not a second independent efficacy test (it followed unblinding to the primary result, so carryover and expectation effects limit how much weight it can carry alone). Dosing was a **single fixed regimen**: 1,500 mg IV every 2 weeks ×3, then every 4 weeks ×4, with **no dose-ranging performed**. That's a real evidentiary gap relative to both ionalumab and iscalimab, whose own Phase 2b programs tested three doses each (see Part V).

PART III · PHASE 2

3.2 Baseline Characteristics

Table 1 from the published paper, reproduced in full below (both populations, both arms). These values were re-verified directly against the source PDF three separate times over the project's history, including once specifically to check four values an external review proposed changing. On direct re-inspection, the review's proposed alternates did not match the source (they appear to reflect a row-transposition error), and I kept the original transcription. One genuine error was found and fixed: the Schirmer's-test criterion direction (corrected to ≤5mm/5min, the clinically correct direction for low tear production).

Characteristic	Pop 1 Placebo (n=38)	Pop 1 DAZ (n=36)	Pop 2 Placebo (n=55)	Pop 2 DAZ (n=54)
Age, years (mean±SD)	48.8±12.1	51.7±9.6	49.0±12.6	50.9±11.7
Female, n (%)	38 (100)	35 (97.2)	50 (90.9)	53 (98.1)
White, n (%)	34 (89.5)	26 (72.2)	31 (56.4)	23 (42.6)
Black/African American, n (%)	1 (2.6)	3 (8.3)	3 (5.5)	4 (7.4)
Asian, n (%)	0	2 (5.6)	5 (9.1)	11 (20.4)
Hispanic or Latino, n (%)	8 (21.1)	8 (22.2)	23 (41.8)	22 (40.7)
Baseline ESSDAI, mean±SD	10.1±4.1	11.4±4.5	2.5±1.6	3.1±1.8
ESSDAI ≥10, n (%)	20 (52.6)	21 (58.3)	0	0
Baseline ESSPRI total, mean±SD	6.6±1.8	6.6±1.6	7.1±1.1	7.5±1.5
— Dryness	6.9±2.3	7.3±1.7	6.8±1.2	7.1±1.6
— Fatigue	6.8±2.2	7.0±1.7	6.8±1.8	7.1±1.8
— Pain	6.1±2.3	5.5±3.0	6.3±1.9	6.6±2.6
OSDI total, mean±SD	40.0±20.9	48.7±25.0	48.6±19.2	48.2±23.8
Schirmer's ≤5mm/5min (≥1 eye), n (%)	31 (81.6)	26 (72.2)	48 (87.3)	46 (86.8)
Positive SS-A, n (%)	37 (97.4)	35 (97.2)	49 (89.1)	51 (94.4)
Positive rheumatoid factor, n (%)	16 (42.1)	15 (41.7)	25 (45.5)	28 (51.9)
Stimulated salivary flow, mL/min	0.97±0.85	0.88±0.64	0.83±0.83	1.10±1.18
Concomitant RA or SLE, n (%)	9 (23.7)	7 (19.4)	7 (12.7)	4 (7.4)

Background therapy at baseline, not shown in the table above: Population 1 was 59.5% on an antimalarial, 41.9% on a glucocorticoid, and 31.1% on a conventional DMARD (arms combined). Population 2's cholinergic-agonist use differed by arm (11.1% dazodalibep vs. 27.3% placebo), a baseline imbalance worth keeping in mind when reading Population 2's placebo-response magnitude.

On the Phase 2 → 3 population bridge: Population 1's Phase 2 eligibility floor (ESSDAI≥5) is *identical* to NCT06104124's. The observed Phase 2 mean (10.1–11.4) simply happened to run above that minimum. This does not establish that Phase 3 will enroll a different average severity; that requires actual information about who Phase 3 enrolled, which is not public. A more concrete, source-backed transportability signal comes from Extended Data Table 1 instead: patients **without** concomitant RA/SLE showed a placebo-adjusted ESSDAI change of ≈−1.6, versus ≈−3.3 among those **with** RA/SLE: a real, if small-sample, clue about what could move Phase 3's effect size relative to Phase 2's 19–24% RA/SLE overlap.

PART III · PHASE 2

3.3 Efficacy Results

Population 1 (systemic, ESSDAI), the primary bridge to NCT06104124:

Endpoint	Dazodalibep	Placebo	Difference	Result
ESSDAI change, Day 169 (LS mean±SE) — primary	−6.3±0.6	−4.1±0.6	−2.2 (90% CI −3.6 to −0.7)	P=0.0167 — significant
ESSDAI(3) responder	72.2%	59.5%	—	P=0.3283 — not significant
ESSDAI(4) responder	66.7%	48.6%	—	P=0.1823 — not significant
ESSDAI(5) responder POST HOC	61.1%	35.1%	—	P=0.0449
ESSPRI change, Day 169 (secondary)	−1.8±0.3	−1.1±0.3	—	P=0.1110 — not significant

Population 2 (symptomatic, ESSPRI), the bridge to NCT06245408, produced the cleanest result in the entire Phase 2 program:

Endpoint	Dazodalibep	Placebo	Difference	Result
ESSPRI total change, Day 169 — primary	−1.80	−0.53	−1.27 (90% CI −1.82 to −0.73)	P=0.0002
— Dryness domain	−1.9	−0.8	—	P=0.0066
— Fatigue domain	−1.7	−0.3	—	P=0.0022
— Pain domain	−1.8	−0.4	—	P=0.0010

All three domains move the same direction as the total score, a useful consistency check, though each domain p-value is unadjusted for multiplicity, so treat them as supportive, not as three additional independently confirmatory results.

Supportive secondary outcomes, Day 169, unadjusted for multiplicity and read as consistency checks rather than confirmatory results on their own:

Outcome	Population	DAZ / Placebo change	P-value
FACIT-Fatigue	Systemic	+8.1 / +5.8	0.3028
FACIT-Fatigue	Symptomatic	+8.1 / +2.8	0.0095
OSDI	Systemic	−16.00 / −14.02	0.6583
OSDI	Symptomatic	−13.95 / −8.52	0.1936
Stimulated salivary flow	Systemic	+0.39 / +0.14	0.1330
Stimulated salivary flow	Symptomatic	+0.20 / −0.01	0.2170

Positive FACIT-Fatigue change = improvement; negative OSDI/ESSPRI change = improvement. None of these six comparisons reaches conventional significance on its own. They're read alongside the primary results above, not instead of them.

A statistical design caveat that applies to every P-value above: the paper explicitly used a relaxed two-sided significance threshold of $\alpha=0.10$ (not the standard 0.05), citing small-N concerns. The primary ESSDAI result clears even the stricter conventional bar, but Population 1's ESSPRI secondary (P=0.1110) does not clear even the paper's own loosened bar. Phase 3's statistical analysis plan will almost certainly use the standard 0.05, so I want to be precise about which Phase 2 results are genuinely robust versus directional-only.

PART III · PHASE 2

3.4 Biomarkers & Crossover Durability

Two blood biomarkers provide evidence of biological activity independent of the (noisier) clinical scores: **CXCL13** (a chemokine marking germinal-center activity) and **rheumatoid factor** (an autoantibody).

Biomarker	Pop 1 adjusted GMR (DAZ/PBO)	Pop 2 adjusted GMR (DAZ/PBO)
CXCL13	0.67 (P=0.0010)	0.56 (P<0.0001)
Rheumatoid factor	0.66 (P<0.0001)	0.54 (P<0.0001)

Both markers suppressed on-drug and **rebounded toward baseline within ≈2 months of stopping drug** during the crossover: direct, drug-attributable evidence, not merely a coincidence of which patients happened to be assigned where. The crossover's clinical-score trajectories point the same direction: Population 1's placebo → DAZ arm improved further after switching (ESSDAI −4.1 → −6.3), while its DAZ → placebo arm partially reversed (−6.3 → −4.4); Population 2's DAZ → placebo arm held essentially steady (ESSPRI −1.8 → −1.9), suggesting more durable symptom benefit than systemic benefit after stopping drug. As throughout this project, the crossover is treated as supportive durability evidence, not a substitute for a true 48-week parallel-group replication.

PART III · PHASE 2

3.5 Safety

Three investigator-attributed, drug-related serious events occurred in two participants, alongside one separately reported death considered unrelated:

Event	Population	Attribution & timing
Deep vein thrombosis + drug-induced liver injury (same participant)	Pop 1, Stage II	Investigator-attributed as related ; onset 180 days post-final-dose
Invasive ductal breast carcinoma	Pop 2, Stage II	Investigator-attributed as related ; onset 139 days post-final-dose
Death, unknown cause following COVID-19 diagnosis	Pop 1	Investigator-considered unrelated; reported during Stage I, tabulated in the Stage II+follow-up safety window (46 days post-last-dose)

Denominators: systemic DAZ n=36 (34 in Stage II), symptomatic safety set n=48. The paper's own defense of the DVT/liver-injury event is narrower than “no signal”: it argues the event's *timing* (180 days post-dose, well outside the expected exposure window) is inconsistent with the platelet Fc-crosslinking mechanism that caused first-generation anti-CD40L thromboembolism. That's a real distinction, but the event itself remains investigator-attributed as drug-related, and a small trial can neither prove nor rule out a real risk from a single event. This is squarely a “watch closely in the Phase 3 safety database” item.

PART IV

Phase 3 Trial Designs & the Phase 2 → 3 Bridge

What the two Phase 3 registry entries actually say, what specifically changes relative to Phase 2, and the pharmacology behind Phase 3's dose selection.

PART IV · PHASE 3 DESIGN

4.1 Registry Comparison

Pulled directly from each trial's ClinicalTrials.gov export.

Field	NCT06104124 (systemic)	NCT06245408 (symptomatic)
Title focus	Moderate-to-severe systemic disease activity	Moderate-to-severe symptom state
Enrollment	651	434
Arms	2 doses + placebo (3 arms)	Dazodalibep + placebo
Primary outcome(s)	ESSDAI change, Week 48	ESSPRI change <i>and</i> DASPRI change, Week 48
Selected secondary outcomes	ESSDAI response proportion; tender/swollen joint counts; PROMIS-Fatigue; ESSDAI at Weeks 12/24; stimulated salivary flow; TEAEs/TESAEs/AESIs to Week 56	DASPRI/ESSPRI meaningful improvement; PROMIS-Fatigue; SF-36 PCS; domain-level DASPRI/ESSPRI (dryness, pain, fatigue); TEAEs/TESAEs/AESIs to Week 56
Status (as of cutoff)	COMPLETED	ACTIVE, NOT RECRUITING
Primary completion	2026-07-23 ACTUAL	2026-10-22 ESTIMATED
Study completion	2026-08-17 ACTUAL	2026-12-17 ESTIMATED
Masking	Quadruple (participant, care provider, investigator, outcomes assessor)	Quadruple (same)

Completion is not disclosure. NCT06104124 finished collecting data in mid-2026, but that says nothing about when Amgen will publicly announce results. No specific announcement date is established or assumed anywhere in this project.

PART IV · PHASE 3 DESIGN

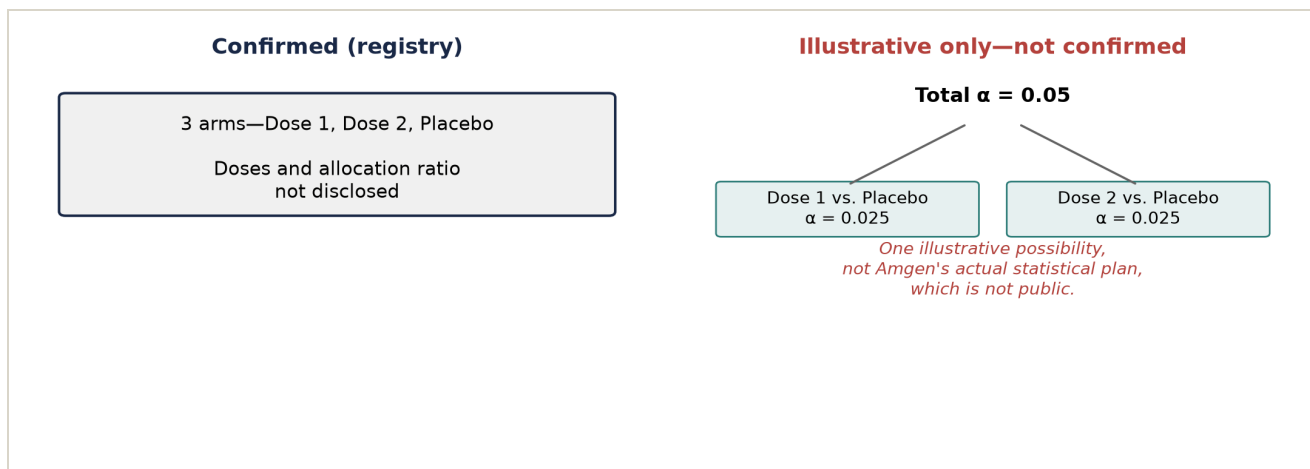
4.2 What Changes Going Into Phase 3

Factor	Phase 2 (systemic)	Phase 3 (systemic)	Implication
Randomized total	74	651	~6.0× per-arm increase (36 → ~217) — more precision, not proportional effect growth
Primary timepoint	Day 169	Week 48 (~day 336)	Longer durability window; more time for placebo response to accumulate
Active regimens	One dose	Two dose arms	Dose selection and multiplicity now matter (see 4.3 and Part VI)
ESSDAI entry floor	≥5	≥5	Unchanged — no loosening established
Baseline/domain mix	Observed	Not established	Real transportability uncertainty

PART IV · PHASE 3 DESIGN

4.3 Dose & Exposure Bridge (PK/PD)

Phase 2's single-dose design (no dose-ranging) is a real evidentiary gap relative to both competitor programs. That gap is **partially, not fully**, closed by a population pharmacokinetic/pharmacodynamic model Amgen conducted across four trials (healthy volunteers, RA patients, and Sjögren's patients).



Key findings: **Sjögren's patients clear the drug more slowly** than healthy volunteers or RA patients (a real, population-specific finding, not a generic assumption), while age, sex, race, anti-drug antibodies, and renal function showed no clinically relevant PK impact. The model related exposure to three immune biomarkers (Ki67+ B-cells, rheumatoid factor, CXCL13) and explicitly states the proposed Phase 3 doses are "expected to achieve sustained average drug concentration above the IC50" of the relevant biomarkers.

What this does and does not resolve: it resolves the concern that Phase 3 doses were chosen with no quantitative basis at all. It does *not* resolve whether either dose produces a *larger clinical* effect than Phase 2's 1,500mg. Sustained biomarker suppression is a pharmacodynamic target, not a clinical efficacy guarantee. The actual milligram doses tested in Phase 3 remain undisclosed in every public source checked. Separately, two active doses against one shared placebo raises a **multiplicity question** with no public testing-hierarchy disclosure, developed fully in Part VI.

PART V

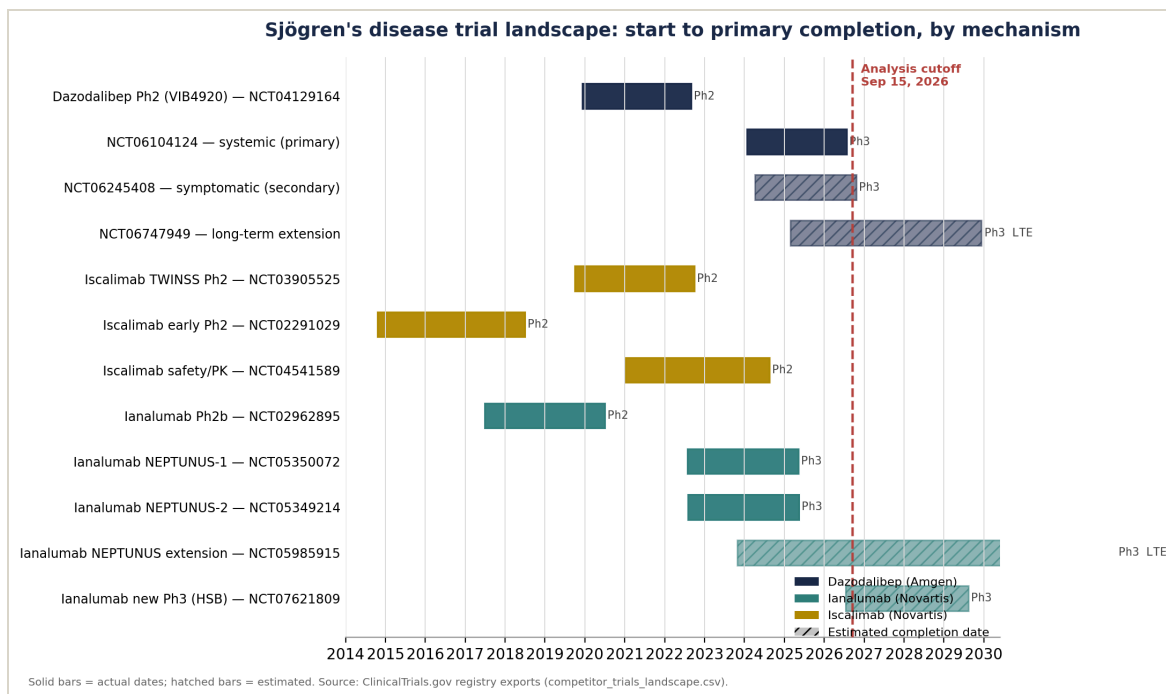
Competitive & Historical Landscape

Every Sjögren's disease trial identified across dazodalibep, ionalumab, and iscalimab (17 registry entries in total), plus the broader historical record across unrelated mechanisms.

PART V · LANDSCAPE

5.1 Full Trial Landscape

The competitive research pulled every Sjögren's disease trial registered for dazodalibep and its two closest competitors: 17 registry entries spanning Phase 1 through long-term extensions, including completed, active, recruiting, and even withdrawn studies. The timeline below places each one in calendar context against this report's analysis cutoff.



A notable, newly-identified data point: Novartis opened a **new** ianalumab Phase 3 trial (**NCT07621809**) in July 2026, specifically enrolling a “high symptom burden” population, conceptually the ianalumab analogue of dazodalibep's own symptomatic trial and a signal that Novartis sees continued commercial logic in the symptom-burden population even after its systemic Phase 3 wins. The full table of all 17 trials, with enrollment, design, and status, is reproduced in Appendix B.

PART V · LANDSCAPE

5.2 Ianalumab (Novartis) Deep Dive

Ianalumab is the most important external precedent in this project: it is the only drug with completed **Phase 3** results on a comparable week-48 ESSDAI endpoint.

Phase 2b (NCT02962895, N=190, doses 5/50/300mg): the trial's actual prespecified primary objective was a **dose-response test** (MCP-Mod on continuous ESSDAI at Week 24): **met**, significant in 4 of 5 models. The individual 300mg-vs-placebo pairwise comparison (a secondary analysis) was not significant (diff -1.92 , $P=0.0921$). An earlier project draft mischaracterized this secondary miss as the trial's primary result, which materially understated ianalumab as a precedent.

Trial	Design	Result
NEPTUNUS-1 (NCT05350072)	2-arm, monthly SC vs. placebo, Wk48, N=275 (137/138)	−6.4 vs. −5.1, diff −1.3 (SE 0.66; 95% CI −2.6 to 0.0), P=0.0496
NEPTUNUS-2, monthly (NCT05349214)	3-arm incl. placebo, Wk48, N=506 (registry) / 504 (conference) DISCREPANCY UNRESOLVED	−6.5 vs. −5.5, diff −1.0, P=0.041
NEPTUNUS-2, every-3-months	Same trial, less-frequent dosing arm	Did not reach significance — a real dose-frequency-sensitivity data point
Pooled (monthly arms only)	Post hoc combination	−6.5 vs. −5.3, P=0.0031 — not a substitute for two individually significant confirmatory trials

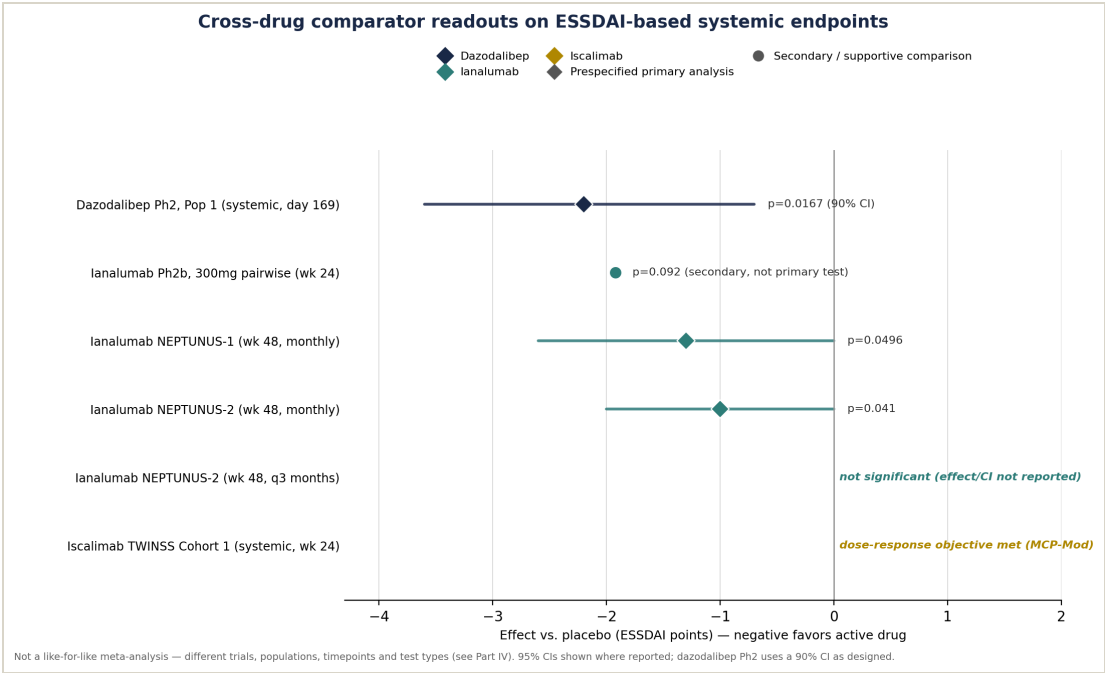
Both individual monthly-arm wins were narrow (p just under 0.05 in each case), a caution against assuming a real effect in this disease/endpoint combination will read out with a comfortable margin. A corrected calculation (an earlier draft claimed the same 1.3-point effect would produce $p=0.0002$ at dazodalibep's scale, using an unrealistically low variance assumption) shows: scaling NEPTUNUS-1's own implied SE for dazodalibep's larger assumed N ($SE \approx 0.66 \times \sqrt{(137/217)} \approx 0.524$), a 1.3-point observed difference implies **$p \approx 0.013$** : still a pass, but a meaningfully less dominant one than first claimed.

PART V · LANDSCAPE

5.3 Iscalimab (Novartis) & the Broader Comparator Set

Drug / trial	Design	Primary result
Iscalimab TWINSS Cohort 1 (systemic, N=173)	Dose-response test, Wk24	Met
Iscalimab TWINSS Cohort 2 (symptomatic, N=100)	600mg vs. placebo, ESSPRI, Wk24	Diff −0.57 (95% CI −1.30 to 0.15), P=0.12 — not significant

Iscalimab's symptomatic-cohort miss is useful, deliberately-sought counterevidence against assuming a class-wide symptom benefit just because the systemic/pathway story works. That's directly relevant to how much conviction the DASPRI risk (Part VII) deserves.



The chart above places every comparable readout on one axis. It is explicitly **not a meta-analysis** (the trials differ in population, timepoint, and test type), but the visual pattern is informative: every CD40-pathway or BAFF-pathway trial with a continuous-ESSDAI or dose-response primary analysis met its primary objective, while margins across the board (ianalumab especially) were narrow rather than dominant.

PART V · LANDSCAPE

5.4 Historical Evidence: Unrelated Mechanisms

A broader historical sweep of Sjögren's trials using unrelated mechanisms, rebuilt as a **qualitative** evidence table after an earlier draft incorrectly computed a numeric “historical base rate” from trials that do not share a common endpoint definition.

Drug	Trial	N	Primary endpoint (exact)	Result
Rituximab	TEARS	~120	≥30% improvement in ≥2 of 4 patient VAS scales	Missed
Rituximab	TRACTISS	133	≥30% reduction, fatigue OR oral-dryness VAS, Wk48 (NOT ESSDAI)	Missed — 39.8% vs. 36.8%
Abatacept	ASAP-III	80	Continuous ESSDAI, Wk24	Missed (open-label extension to Wk48 showed improvement, but uncontrolled)
Tocilizumab	ETAP	110	Composite: ESSDAI≥3pt AND no new domain activity AND no PGA worsening	Missed — 52.7% vs. 63.6% PLACEBO NUMERICALLY HIGHER
Belimumab	BELISS	30	Open-label, no placebo arm	Not a controlled comparison — excluded from any tally

What this table does not support: a computed historical “base rate” or probability input. These trials are not a systematic sample, do not share one endpoint definition (VAS composite, continuous ESSDAI, composite responder, dose-response test, and open-label are all different statistical questions), and are too few to support a numeric rate. It is also improper to count dazodalibep's own Phase 2 result both as an independent data point in this table *and* separately as evidence for its own forecast: that double-counts the same evidence. The table is retained here as qualitative context only: a reminder that autoimmune Sjögren's trials fail often enough, for varied and endpoint-specific reasons, that a positive call should never be treated as a formality.

PART VI

Statistical Scenario Modeling

The quantitative core of the forecast: variance sensitivity, bull/base/bear scenarios, and why the multiplicity question keeps the true significance bar uncertain.

PART VI · MODELING

6.1 Three Distinct Quantities

An early modeling draft conflated three genuinely different statistical quantities. Keeping them separate is the single most important discipline in this section:

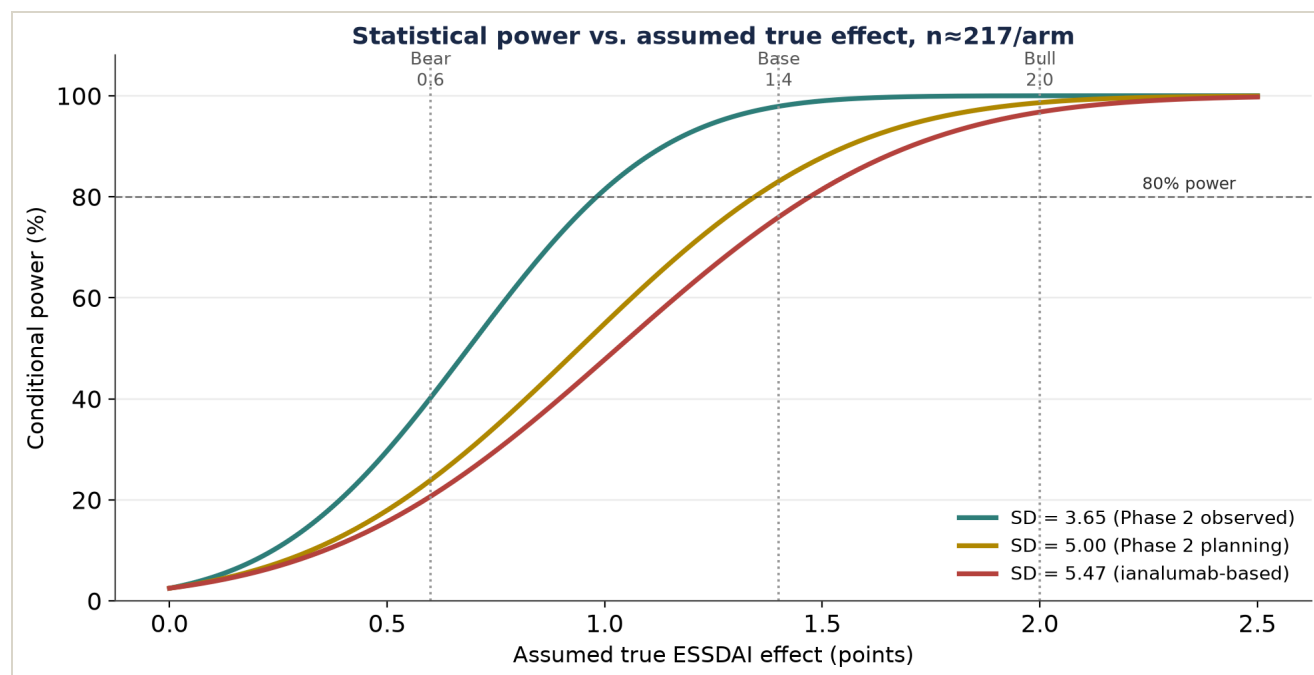
1. **Observed significance threshold:** the delta needed to be statistically significant in one trial, given its size and a variance assumption.
2. **Effect associated with a given power level:** the true delta that would produce an 80% probability of significance *if the trial were repeated many times*. This is a property of the design, not a prediction of any single outcome.
3. **Predictive probability of trial success:** requires integrating over uncertainty in the true effect, true variance, dose performance, missing data, and multiplicity correction. This report does not compute this number, because the inputs needed to do so honestly are not publicly available.

A single point estimate of “the true delta is X” does not exist before a readout. Everything in this section is conditional on stated assumptions, not a forecast of what will happen.

PART VI · MODELING

6.2 Variance Sensitivity

The central correction to this project's statistics: an early model used only the single most favorable variance assumption (SD≈3.65, back-calculated from dazodalibep's own Phase 2 standard errors) without testing sensitivity to two other directly relevant assumptions: the Phase 2 protocol's own planning SD (5.00), and the effective SD implied by NEPTUNUS-1's reported variance (≈5.47), the only real external Phase 3 benchmark available. None of the three is the actual, undisclosed dazodalibep Phase 3 variance.



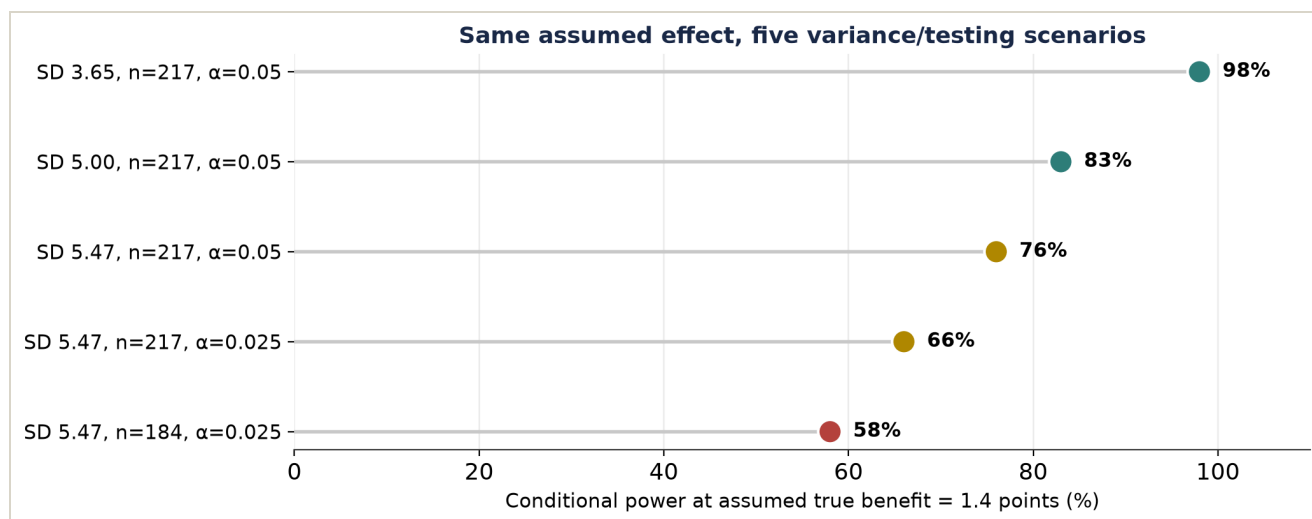
Variance assumption	Per-arm n	Two-sided α	Effect for 80% power	Conditional power if true benefit = 1.4
Original model, SD 3.65	217	0.05	0.98	98%
Phase 2 planning assumption, SD 5.00	217	0.05	1.34	83%
Competitor-based, SD 5.47 (NEPTUNUS-1)	217	0.05	1.47	76%
Same, illustrative stricter α (multiplicity)	217	0.025	1.62	66%
Same, illustrative missing-data/information loss	184	0.025	1.76	58%

At the SD=3.65, $\alpha=0.05$ baseline, the bare **observed-significance threshold** (a different quantity from the 80%-power effect) is ≈ 0.69 , not the 1.0 figure loosely used as “the minimum detectable effect” in earlier drafts. A true effect of 0.7 does not guarantee a pass under this assumption; it produces roughly a coin-flip probability in any single trial.

PART VI · MODELING

6.3 Bull / Base / Bear Scenarios

Holding variance at the externally-anchored SD=5.47, three assumed-true-effect scenarios were tested against Phase 2's own observed 2.2-point effect as a reference point (not a fourth scenario).



Scenario	Assumed true delta	Retention vs. Ph2's 2.2	Conditional power	Read
Bull	2.0	91%	$\approx 97\%$	Would very likely pass if true
Base	1.4	64%	$\approx 76\%$	Would likely pass, real but not enormous margin
Bear	0.6	27%	$\approx 21\%$	Would likely not reach significance if true

These are **assumed inputs, not predicted outcomes**, and the base case (1.4) is explicitly a judgment call, not a fitted or empirically estimated number. Illustrative confidence weights were assigned to each scenario (Bull $\approx 20\%$ / Base $\approx 50\%$ / Bear $\approx 30\%$) but are labeled throughout as analyst judgment, not an empirically validated probability distribution, and are not presented anywhere in this project as a calibrated forecast.

What the single available real-world transition (ianalumab) can and cannot tell us: ianalumab's own per-arm scale-up from Ph2b to Phase 3 was $\approx 2.9\times$ ($47 \rightarrow 137$ per arm); dazodalibep's assumed scale-up is $\approx 6.0\times$ ($36 \rightarrow \sim 217$): roughly twice as large a jump, with no precedent for how much effect retention survives at that scale. This should be read as a qualitative sanity check, not a quantitative multiplier applied with false precision.

PART VI · MODELING

6.4 Multiplicity & the Actual Success Rule

NCT06104124 tests **two active dose arms against one shared placebo**. This raises a family-wise error question that the public registry does not resolve: whether Amgen uses a hierarchical gatekeeping procedure, a fixed alpha split, or some other multiplicity control is not disclosed. The illustrative $\alpha=0.025$ row used throughout this section (an even split of a 0.05 budget across two doses) is **one simple possibility, not an assertion about Amgen's actual statistical plan**.

Any single “the trial will pass/fail” statement built on this project's modeling needs to carry this caveat explicitly: the real, multiplicity-adjusted significance bar could be meaningfully stricter than a naive single-comparison 0.05 threshold, depending on an undisclosed testing hierarchy.

PART VII

Risk Assessment

The four material, disclosed risks to the forecast, not softened into a generic disclaimer but named, scoped, and weighed individually.

PART VII · RISKS

7.1 The DASPRI Risk (Secondary Trial)

NCT06245408's registered primary outcomes are **both** ESSPRI change **and** DASPRI change at Week 48. An early project draft discussed only ESSPRI, a material gap given DASPRI is a meaningfully different measurement approach (a repeated/recall-based diary, versus a single-visit questionnaire).

Question	Evidence	Interpretation
Does DAZ improve reported symptoms?	Positive Phase 2 ESSPRI (P=0.0002, the cleanest result in the program)	Supports a positive ESSPRI lean
What supports DASPRI specifically?	Nothing — a full-text search of the Phase 2 paper confirms zero occurrences	No dazodalibep-specific precedent exists for this instrument at all
What defines overall trial success?	Two co-primary outcomes; the pass/fail rule (both required, either sufficient, or a hierarchy) is not disclosed	Do not assume either interpretation

DASPRI is a newer instrument developed to align with FDA patient-reported-outcome guidance, plausibly added for regulatory-alignment reasons, not because ESSPRI underperformed. That context matters, but the practical fact remains: **dazodalibep has never been tested on this instrument**, so no amount of re-reading the Phase 2 paper produces evidence that doesn't exist. A supportive, non-identical precedent exists in the JOQUER trial (hydroxychloroquine, N=120): its overall ESSPRI result was null, but a post hoc “high symptom burden” subgroup (n=32) showed a relative improvement: a phenotype conceptually similar to, but not procedurally identical to, NCT06245408's own ESSPRI $\geq$ 5 enrichment rule.

PART VII · RISKS

7.2 Safety Risk

Summarized fully in Part III.5: three investigator-attributed, drug-related serious events in two Phase 2 participants (a DVT/liver-injury pair and a breast carcinoma), plus one separately reported unrelated death. The Fc-free engineering specifically targets the known first-generation anti-CD40L thromboembolism mechanism (a real, quantifiable de-risking step), but does not eliminate thrombotic risk from other causes categorically, and a trial this small can neither confirm nor rule out a real signal from individual events. This is an open item for the (much larger) Phase 3 safety database to resolve, and a reminder that a statistically positive efficacy result and an acceptable overall benefit-risk profile are separate questions (developed further in Part VIII.4).

PART VII · RISKS

7.3 Dose Selection Risk

Covered in Part IV.3: no empirical dose-ranging was performed in Phase 2, and the actual Phase 3 doses remain undisclosed. The PK/PD model partially mitigates this risk by grounding dose selection in target biomarker coverage, but that is a pharmacodynamic justification, not a guarantee about the resulting clinical (ESSDAI) effect size at either dose.

PART VII · RISKS**7.4 Population Transportability Risk**

Whether Phase 3's much larger, more geographically dispersed population resembles Phase 2's small, more tightly observed cohort closely enough for the observed effect to transfer is a genuinely open question. It isn't resolved by eligibility criteria alone, since both trials share the identical ESSDAI $\geq$ 5 floor. The one source-backed clue available (the RA/SLE-overlap subgroup difference, Part III.2) is suggestive, not conclusive, given small subgroup sizes.

PART VIII

Final Thesis & Forecast

The project's headline call for NCT06104124, its single strongest counter-argument, and a framework for interpreting whatever the actual readout shows.

PART VIII · THESIS

8.1 **Headline Call & Ranked Evidence**

We lean positive on NCT06104124's ESSDAI primary endpoint, with material uncertainty that should not be understated. The trial's large enrollment means it does not need to fully replicate Phase 2's observed effect to reach significance, but how much smaller an effect it can tolerate depends heavily on which variance assumption is used, and the true variance for this specific Phase 3 trial is not publicly known. This is a **defensible lean, not a high-confidence prediction.**

Supporting evidence, ranked:

1. **Phase 2's randomized primary result is real and appropriately powered for its own size:** ESSDAI delta -2.2 , $P=0.0167$, clearing even the standard 0.05 bar despite the trial's own relaxed 0.10 threshold.
2. **Pharmacodynamic evidence supports real target engagement,** separate from and more reliable than the clinical scores: CXCL13 and rheumatoid factor both suppressed on-drug, rebounded off-drug.
3. **External evidence this disease and pathway can respond at Phase 3 scale:** ionalumab won both individual Phase 3 monthly-arm comparisons, narrowly but really, removing “no ESSDAI-based Phase 3 has ever succeeded” as a reason for doubt.
4. **A qualitative, not statistically-weighted, pattern:** both CD40-pathway trials with a comparable primary analysis (dazodalibep's continuous-ESSDAI test, iscalimab's dose-response test) met their primary objective.

PART VIII · THESIS

8.2 **Strongest Counter-Argument**

The variance assumption that made this trial's cushion look large was the most optimistic of at least three plausible ones, and the real external benchmark points to a meaningfully larger variance. Under dazodalibep's own Phase 2 SEs ($SD \approx 3.65$), the trial looks comfortably powered; even a 1.4-point effect would appear highly significant. But the Phase 2 protocol's own planning assumption used $SD=5.00$, and the only real Phase 3 benchmark available (NEPTUNUS-1) implies an effective $SD \approx 5.47$. Under that more conservative, externally-anchored variance, a true 1.4-point effect drops from 98% conditional power to **76%**, still favorable, but a meaningfully different confidence level, and before accounting for whatever multiplicity correction the two active dose arms actually require.

Reinforcing this: ionalumab's own two individual Phase 3 wins ($p=0.0496$, $p=0.041$) were both narrow. The one available real-world precedent for “a real effect in this exact disease/endpoint combination” has not produced a comfortable margin.

PART VIII · THESIS

8.3 **Clinical Meaning: Three Separate Questions**

A “positive” call under this project's modeling answers only the first of three distinct questions that are easy to conflate:

Question	What it requires	What this report forecasts
1. Statistical significance	Will the primary ESSDAI comparison clear its (multiplicity-adjusted) threshold?	Yes, this is what Parts VI–VIII forecast — a necessary condition for a conventional “win,” and the literal question the assignment brief asks
2. Convincing patient-level benefit	Is the magnitude large enough to matter per patient (the within-patient MCII is ≥ 3 ESSDAI points)?	Not addressed — ionalumab's own Phase 3 wins (1.0–1.3 points) were themselves below this individual threshold, a pattern that could repeat here
3. Acceptable overall benefit-risk profile	Does efficacy outweigh the Phase 2 safety signal, in an unmet-need context?	Not addressed by the statistical forecast — a separate judgment

A statistically positive call should not be silently overstated into satisfying questions 2 or 3. The honest framing carried throughout this project is that the trial is more likely than not to be *statistically* positive, and that result would still need separate evaluation for clinical magnitude and safety trade-off before calling the drug “successful” in the broader sense a physician or regulator would use.

PART VIII • THESIS

8.4 Readout Interpretation Matrix

A prespecified framework for interpreting whatever the actual readout shows, across five dimensions, built specifically so the call can be evaluated honestly against reality rather than redefined after the fact.

Dimension	Would strengthen	Would weaken	Would reverse
Efficacy (statistical)	Delta ≥ 1.4 , $p < 0.01$, both doses consistent	Delta 0.7–1.0, p near 0.05 (ionalumab-like)	Delta < 0.7 , non-significant primary
Efficacy (clinical meaning)	Responder analyses (≥ 3 pt) also significant	Continuous score passes but responders miss again	Any dose arm trends worse than placebo
Dose/exposure	Disclosed exposure exceeds Phase 2's, consistent with PK/PD target	No dose rationale disclosed	Evidence dose deviated from the PK/PD-modeled target
Secondary trial (NCT06245408)	Clean positive ESSPRI/DASPRI readout	Narrow or mixed result	Clearly negative, given the shared molecule
Safety	No new signals beyond Phase 2's	Modest increase in infection/thromboembolic events	A clear, replicated thromboembolic signal above background

BOTTOM LINE

Is the Forecast Positive or Negative?

This section states plainly what the entire project converges on: is the forecast positive or negative, and why.

OVERALL CALL

Net **POSITIVE lean** on NCT06104124's ESSDAI primary endpoint: a defensible, evidence-backed forecast carrying **material, quantified residual risk**, not a high-confidence prediction.

Net **positive lean, with a specific disclosed gap** (DASPRI) on NCT06245408's co-primary endpoints: directionally favorable on the evidence that exists, unresolved on the evidence that does not yet exist.

Why the call is positive, not neutral or negative

Three independent lines of evidence point the same direction: a statistically real and appropriately powered Phase 2 result (−2.2 ESSDAI points, $P=0.0167$, clearing even the conventional 0.05 bar despite the trial's own relaxed 0.10 standard); direct pharmacodynamic proof of biological activity (CXCL13 and rheumatoid factor suppression, reversible on stopping drug); and external confirmation that a comparable endpoint *can* be won at Phase 3 scale in this exact disease (ianalumab's two individually significant NEPTUNUS wins). None of these three pillars depends on the others being true, which is what makes the positive lean more than a single fragile data point.

Why the risk is real, not a hedge

The same modeling that supports the positive lean also quantifies exactly how it could fail: under the most defensible externally-anchored variance assumption ($SD=5.47$, the only real comparable Phase 3 variance available), a plausible base-case true effect (1.4 points, 64% of Phase 2's observed effect) carries only **76% conditional power**, meaningfully short of certainty, and that's before accounting for whatever multiplicity correction the trial's two active dose arms actually require, which could push the effective bar higher still. Ianalumab's own two Phase 3 wins were both narrow, not dominant. The one real precedent available argues for humility about margin, not just direction.

Why the symptomatic trial gets a similar but distinctly-reasoned call

NCT06245408's own Phase 2 precursor produced the single cleanest result in the entire program ($P=0.0002$), which argues for real optimism on the ESSPRI component specifically. What caps full conviction is not weak evidence but a **named, specific gap**: DASPRI, a co-primary endpoint with literally zero dazodalibep-specific precedent and an undisclosed pass/fail rule relative to ESSPRI. This is a fact about the evidence base that no further analysis of existing data can resolve. Only the readout itself can.

What would change this call

- Disclosure of the actual Phase 3 statistical analysis plan (variance assumption, multiplicity control, testing hierarchy): this would replace every sensitivity range in this report with real numbers.
- Any information on Phase 3's actual enrolled baseline ESSDAI distribution or RA/SLE overlap.
- A negative or narrow NCT06245408 readout, given the shared molecule and mechanism.

- Resolution of the NEPTUNUS-2 N=506-vs-504 discrepancy and similar small data-reconciliation gaps noted throughout.

Bottom line

Weighing the evidence and the risks together, my read is that dazodalibep is **more likely than not** to deliver a statistically positive systemic ESSDAI result in NCT06104124, on the strength of a genuinely strong, biologically coherent Phase 2 signal and real external precedent that the endpoint is achievable at Phase 3 scale. But this is a probabilistic lean under quantified uncertainty (roughly two-thirds to three-quarters conditional power under this project's own base-case and sensitivity assumptions), not a near-certain outcome. A negative primary result would not be a shocking surprise given how sensitive the power calculations are to unknowns that remain genuinely unknown as of this report's analysis cutoff. Both calls should be read as informed, evidence-based forecasts under real uncertainty, exactly the standard this project held itself to throughout, including when that standard required correcting its own earlier, more confident conclusions.

REFERENCE

Appendix A Master Evidence Table

Every material number used anywhere in this report traces back to one row in this table: claim, number, population, timepoint, analysis type, source, and the caveat that qualifies it. This is the project's single source of truth, built per the external reviewer's recommendation so that a number is corrected in one place, not hunted down across a dozen files.

Dazodalibep Phase 2 (St. Clair et al., *Nature Medicine* 2024)

Claim	Number	Population / timepoint	Analysis	Caveat
Primary endpoint, systemic	ESSDAI -6.3 ± 0.6 (DAZ) vs -4.1 ± 0.6 (PBO); diff -2.2 (90% CI $-3.6, -0.7$), $P=0.0167$	Pop 1, $n=74$; Day 169	MMRM, prespecified primary, $\alpha=0.10$ (2-sided)	Small N; relaxed α threshold
Primary endpoint, symptomatic	ESSPRI -1.80 vs -0.53 ; diff -1.27 (90% CI $-1.82, -0.73$), $P=0.0002$	Pop 2, $n=109$; Day 169	MMRM, prespecified primary	Small N
ESSDAI(5) responder	61.1% DAZ vs 35.1% PBO, $P=0.0449$	Pop 1; Day 169	Post hoc	Not prespecified
CXCL13 / RF suppression	Adj. GMR vs PBO: CXCL13 $0.67-0.56$; RF $0.66-0.54$ (all $P \leq 0.001$)	Both populations; Day 169	Exploratory biomarker	Not efficacy proof on its own
RA/SLE overlap subgroup	Placebo-adj. Δ ESSDAI ≈ -1.6 (no RA/SLE) vs ≈ -3.3 (with RA/SLE)	Pop 1 subgroup; Day 169	Exploratory subgroup	Small subgroup N; not a robust interaction claim
Death	$n=1$	Pop 1, Stage II+FU	Safety	Investigator: unrelated to study drug
DVT + drug-induced liver injury	Same participant, both events, onset 180 days post-final dose	Pop 1, Stage II	SAE / AESI	Investigator-attributed as related
Invasive ductal breast carcinoma	$n=1$, onset 139 days post-final dose	Pop 2, Stage II	SAE / AESI	Investigator-attributed as related
Molecular description	Tn3 scaffold, non-antibody, Fc-free fusion protein	—	Design / MOA	Not "PASylated" — corrected error

NCT06104124 (Phase 3, primary trial) & NCT06245408 (Phase 3, secondary trial)

Claim	NCT06104124	NCT06245408	Caveat
Status	COMPLETED (primary completion 2026-07-23, actual)	ACTIVE_NOT_RECRUITING (primary completion 2026-10-22, estimated)	Results not yet posted for either as of the analysis cutoff
Enrollment	$N=651$, 3 arms (2 doses + placebo)	$N=434$	Per-arm split not disclosed
Primary endpoint(s)	ESSDAI change from baseline, Week 48	ESSPRI change AND DASPRI change (both), Week 48	SAP / pass-fail hierarchy not public for either
Eligibility floor	ESSDAI ≥ 5 , anti-Ro/RF+	ESSPRI ≥ 5 , ESSDAI < 5 , salivary flow ≥ 0.1 mL/min	Same floor as respective Phase 2 populations — not looser

Ianalumab (Novartis)

Claim	Number	Analysis	Caveat
Ph2b primary objective	Dose-response MET — significant in 4/5 models	Prespecified MCP-Mod, N=190, Week 24	This is the actual primary result — corrected from an earlier “missed” claim
Ph2b pairwise (secondary)	300mg vs PBO: diff -1.92, P=0.0921	Secondary / supportive	Not significant, and not the primary test
NEPTUNUS-1 primary	-6.4 vs -5.1, diff -1.3 (SE 0.66), P=0.0496	Prespecified primary, N=275, Week 48	Narrow margin
NEPTUNUS-2 primary (monthly)	-6.5 vs -5.5, diff -1.0, P=0.041	Prespecified primary, N=506/504, Week 48	N discrepancy unresolved between registry and conference figures
NEPTUNUS-2 primary (q3-months)	Did not reach significance	Prespecified primary, Week 48	Real dose-frequency-sensitivity data point

Iscalimab (Novartis) & historical comparators

Claim	Number	Caveat
TWINSS Cohort 1 primary	Dose-response MET, N=173 (44/43/43/43), Week 24	Corrected — earlier draft used N=273 (combined total) for this cohort
TWINSS Cohort 2 primary	ESSPRI diff -0.57 (95% CI -1.30, 0.15), P=0.12	Not significant — useful counterevidence to an automatic class-wide symptom benefit
TRACTISS primary	39.8% (rituximab) vs 36.8% (PBO), not significant, N=133	Corrected classification — not an ESSDAI-primary trial (fatigue/dryness VAS composite)
ETAP primary	52.7% vs 63.6%, P=0.14, N=110	Composite response, not continuous ESSDAI; placebo numerically higher
ASAP-III primary	No significant difference, N=80	Open-label extension improvement is uncontrolled, not proof of a missed effect

Statistical model parameters

Claim	Number	Caveat
SD assumption A	3.65	Backed out from dazodalibep Ph2 SEs — most favorable of three, not the actual Ph3 variance
SD assumption B	5.00	Dazodalibep Ph2 protocol's own planning assumption
SD assumption C	5.47	Backed out from NEPTUNUS-1's reported SE≈0.66 — only real external Ph3 benchmark available
MDE at 80% power	0.98 (SD 3.65) / 1.34 (SD 5.00) / 1.47 (SD 5.47)	n=217/arm assumed; per-arm N not confirmed by registry

Open items not yet independently verified: the NEPTUNUS-2 N=506-vs-504 discrepancy; DASPRI results for dazodalibep Phase 2 Population 2 (mentioned in the paper's secondary endpoints but not pulled into this project's files); the actual Phase 3 statistical analysis plan; and the Population 2 fatigue placebo value (6.8 vs 6.9), which is low-confidence either direction on direct re-reading of Table 1. Source: 03_analysis/evidence_table.md.

REFERENCE

Appendix B Full Competitive Trial Landscape (17 Trials)

Every dazodalibep, ionalumab and iscalimab registry entry pulled for this project, not just the trials with disclosed efficacy results. The two withdrawn/zero-enrollment studies and the PK/PD-only studies are included for completeness; they carry no efficacy signal but establish the full development footprint of each molecule.

(A) = actual date, (E) = estimated date, as flagged in the registry export.

NCT	Drug	Phase	Role	Sponsor	Status	N	Start	Primary completion
NCT06104124	Dazodalibep	Ph3	Systemic (primary)	Amgen	COMPLETED	651	2024-01-24	2026-07-23 (A)
NCT06245408	Dazodalibep	Ph3	Symptomatic (secondary)	Amgen	ACTIVE_NOT_RECRUITING	434	2024-04-08	2026-10-22 (E)
NCT06747949	Dazodalibep	Ph3 LTE	Long-term extension	Amgen	RECRUITING	844	2025-02-25	2029-12-11 (E)
NCT04129164	Dazodalibep (VIB4920)	Ph2	Original registrational Ph2	Amgen	COMPLETED	183	2019-12-09	2022-09-01 (A)
NCT05350072	lanalumab	Ph3	NEPTUNUS-1	Novartis	COMPLETED	275	2022-07-28	2025-05-07 (A)
NCT05349214	lanalumab	Ph3	NEPTUNUS-2	Novartis	ACTIVE_NOT_RECRUITING	506	2022-08-04	2025-05-13 (A)
NCT05985915	lanalumab	Ph3 LTE	NEPTUNUS extension	Novartis	ACTIVE_NOT_RECRUITING	612	2023-10-27	2033-04-22 (E)
NCT07621809	lanalumab	Ph3	New high-symptom-burden trial	Novartis	RECRUITING	570	2026-07-16	2029-08-14 (E)
NCT02962895	lanalumab	Ph2b	Dose-response (Bowman 2021)	Novartis	COMPLETED	190	2017-06-27	2020-06-30 (A)
NCT02149420	lanalumab	Ph2	Early PD study	Novartis	COMPLETED	27	2014-05-23	2018-02-07 (A)
NCT05124925	lanalumab	Ph2	Salivary gland biopsy MOA study	Novartis	COMPLETED	21	2022-07-27	2024-06-26 (A)
NCT06293365	lanalumab	Ph2	PK comparability (device switch)	Novartis	ACTIVE_NOT_RECRUITING	155	2024-07-02	2026-02-17 (A)
NCT03905525	Iscalelimab	Ph2	TWINSS (Cohorts 1&2)	Novartis	COMPLETED	273	2019-10-01	2022-09-28 (A)
NCT02291029	Iscalelimab	Ph2	Early safety/PK/efficacy	Novartis	COMPLETED	69	2014-10-22	2018-06-29 (A)
NCT04541589	Iscalelimab	Ph2	Safety/tolerability follow-on	Novartis	COMPLETED	206	2021-01-05	2024-08-19 (A)
NCT02495129	lanalumab	Ph1/2	PET/CT biopsy study	Novartis	WITHDRAWN	0	2015-12	2017-09 (A)
NCT06411639	lanalumab	Ph1	China PK bridging study	Novartis	WITHDRAWN	0	2024-11-25	2025-12-17 (E)

Source: 01_data/competitor_trials_landscape.csv, a 17-row ClinicalTrials.gov registry export covering all three molecules across every phase and role identified during the project's research. See Part V for the narrative discussion and the visual timeline of

the 12 trials with disclosed or estimated completion dates.

METHODOLOGY

Appendix C Correction Log: The External Review Pass

On September 15, 2026, the project's working files were reviewed by an independent pass whose job was to find every place an overconfident or simply incorrect claim had crept into the analysis. That review is treated here as a first-class part of the project's methodology, not a footnote. This appendix documents what was found and what changed as a direct result, in the reviewer's own priority order.

Why this appendix exists. A research process that never shows its corrections looks more confident than it should. The single biggest lesson of this project was that repeating a claim across five summary documents does not make it true. A rigorous forecast has to survive an adversarial re-read of its own sources, not just an internally consistent narrative.

Priority 1: Trial outcomes and the historical comparison

Finding: the project had stated that ionalumab's Phase 2b trial “failed its primary endpoint,” based on the non-significant 300 mg-vs-placebo pairwise comparison ($P=0.092$). In fact the trial's actual prespecified primary objective was a dose-response test (MCP-Mod), which was met in 4 of 5 statistical models. **Correction:** every downstream document describing ionalumab as a failed Phase 2 was rewritten; the “33% historical base rate” and “100% CD40-pathway success rate” figures were removed entirely as invalid probability inputs (the underlying trial set is neither systematic nor endpoint-uniform), and replaced with the qualitative evidence table in Appendix A. TRACTISS and ETAP were reclassified: TRACTISS's true primary endpoint was a fatigue/oral-dryness VAS composite, not ESSDAI; ETAP used a multi-component composite response definition, not continuous ESSDAI.

Priority 2: The Phase 2-to-3 population bridge

Finding: the project had argued Phase 3 eligibility was “looser” than Phase 2 by comparing a Phase 2 observed mean against a Phase 3 eligibility minimum, not a valid comparison, since both trials in fact use the same eligibility floor ($\text{ESSDAI} \geq 5$ for the systemic population). **Correction:** that argument was removed from the baseline memo, the plain-language summary and the scenario model's reasoning. In its place, the paper's own exploratory RA/SLE-overlap subgroup finding was added as a more defensible, if still limited, transportability question (Part VII.4).

Priority 3: The statistical interpretation

Finding: the project's power/significance tables were arithmetically consistent with the chosen inputs, but the presentation blurred three distinct statistical quantities together: the observed-significance threshold, the effect size associated with 80% power, and the predictive probability of trial success. It implied a bigger statistical cushion than a single optimistic variance assumption ($\text{SD}=3.65$) justified. **Correction:** Part VI now explicitly separates these three quantities, presents power under three variance assumptions ($\text{SD } 3.65 / 5.00 / 5.47$) rather than one, and states plainly that a true effect of 0.7 points produces roughly a coin-flip outcome under the model, not a guaranteed pass. The claim that scaling ionalumab's effect to dazodalibep's larger sample size would produce $P=0.0002$ was corrected: the proper sample-size-only rescaling of NEPTUNUS-1's SE ($0.66 \times \sqrt{(137/217)} \approx 0.524$) implies $P \approx 0.013$, not 0.0002. The earlier figure additionally assumed a lower variance than NEPTUNUS-1's own data support.

Priority 4: DASPRI in the symptomatic trial

Finding: NCT06245408's own registry export lists *both* ESSPRI and DASPRI change at Week 48 as co-primary outcomes; the project's narrative discussed only ESSPRI. **Correction:** DASPRI is now discussed explicitly

throughout (Parts II.3, IV.1 and VII.1) as a named, unresolved risk: dazodalibep has zero prior DASPRI-specific data, and the pass/fail hierarchy between the two co-primaries is not publicly disclosed.

Priority 5: Baseline table transcription

Finding: the reviewer flagged six Population 2 baseline values (ESSPRI total, dryness, fatigue, OSDI placebo, and salivary-flow placebo) as possible transcription errors, plus the direction of the Schirmer's-test criterion. **What actually happened next matters as much as the flag itself:** I checked each proposed value directly against the rendered Table 1 image in the source PDF, rather than just accepting it. Only one of the seven flags held up: the Schirmer's-test criterion had indeed been inverted (corrected to “ ≤ 5 mm/5 min in at least one eye,” the clinically correct direction for low tear production). The other six proposed replacement numbers were checked against the source three separate times across this project's history and did *not* match it. They appear to trace to a row-transposition error, repeated identically across two independent review passes. The project's original transcription was correct and was kept. Part III.2 carries the full table with this provenance note attached; the same values also appear in the final submission deck's own baseline slide, so the two documents agree on every figure in that table.

Priority 6: Balancing biology and safety interpretation

Finding: biomarker suppression (CXCL13, RF) had in places been treated as effectively proving clinical efficacy, and the safety summary described “two deaths/serious infections” in language the underlying paper did not support. The paper's Table 3 reports one death (unrelated per investigator) and, separately, an investigator-attributed invasive ductal breast carcinoma AESI. The project had also used the drug's Tn3-scaffold description incorrectly as “PASylated.” **Correction:** Part III.5 now states the safety findings exactly as the paper reports them, with attribution and timing but no added causal inference, and the molecular description was corrected throughout.

Priority 7: Document system and timeline consistency

Finding: the two Phase 3 trials had been described in places as “both finished” and “both using ESSDAI,” which contradicts the underlying registry exports (NCT06104124 is COMPLETED as of 2026-07-23 actual; NCT06245408 is ACTIVE_NOT_RECRUITING with an estimated 2026-10-22 primary completion). **Correction:** every reference to trial status and dates across this report distinguishes actual from estimated dates and treats primary completion, study completion and results-disclosure as three separate milestones that need not occur in the same order for both trials. The master evidence table (Appendix A) was created as this report's single source of truth precisely to prevent this kind of drift from recurring.

Eight additional topics folded in

Beyond the seven corrections above, the review flagged eight topics as worth adding to strengthen the analysis rather than merely fix it: the dose and exposure bridge (Part IV.3), the ESSDAI/ESSPRI/DASPRI testing architecture (Part II.3), missing data and rescue medication considerations (noted throughout Part IV), domain composition and population transportability (Part VII.4), separating the two NEPTUNUS trials rather than reporting a pooled figure (Part V.2), iscalimab's symptom-cohort counterevidence (Part V.3), the distinct clinical meaning of a statistically positive result (Part VIII.3), and a readout interpretation matrix (Part VIII.4). All eight are incorporated into the relevant sections of this report rather than treated as a separate bolt-on.

Source: 00_reference/external_review.md, the full independent review document.

REFERENCE

Appendix D

Glossary

ESSDAI

EULAR Sjögren's Syndrome Disease Activity Index: a clinician-scored measure of systemic (organ-level) disease activity across 12 domains. Lower is better. The primary endpoint of NCT06104124.

ESSPRI

EULAR Sjögren's Syndrome Patient Reported Index: a 3-item patient-reported measure of dryness, fatigue and pain (0–10 each). Lower is better. A primary endpoint of NCT06245408.

DASPRI

Diary for Assessing Sjögren's Patient-Reported Outcome Index: a daily patient diary instrument. Co-primary with ESSPRI in NCT06245408; no prior dazodalibep-specific data exists for it.

MMRM

Mixed Model for Repeated Measures: the statistical model used to analyze the Phase 2 primary endpoints, which handles repeated within-patient measurements and (under standard assumptions) missing data.

MCP-Mod

Multiple Comparison Procedure – Modeling: a prespecified statistical method for testing whether a dose-response relationship exists across multiple candidate dose-response curve shapes. The actual primary analysis method for both ionalumab's Ph2b trial and iscalimab's TWINSS Cohort 1.

MDE

Minimum Detectable Effect: the smallest true treatment effect that a trial has a stated probability (typically 80%) of detecting as statistically significant, given its sample size and assumed variance.

Conditional power

The probability that a trial will reach statistical significance, calculated conditional on an assumed true effect size and variance. Distinct from the observed-significance threshold and from an unconditional predictive probability of success.

CD40/CD40L pathway

A costimulatory signaling pathway between T cells (CD40L) and B cells/antigen-presenting cells (CD40) central to the adaptive immune response implicated in Sjögren's disease. Dazodalibep and iscalimab block this pathway from different sides; ionalumab targets a related but distinct B-cell survival pathway (BAFF-R).

Tn3 scaffold

A non-antibody, Fc-free fusion protein platform used for dazodalibep. Corrected in this report from an earlier, inaccurate “PASylated” description.

AESI

Adverse Event of Special Interest: a predefined category of adverse event that a trial's safety monitoring plan tracks with particular attention, regardless of whether it is deemed drug-related.

Primary completion vs. study completion

Primary completion is the date the last participant was evaluated for the primary endpoint; study completion is the date the entire trial (including long-term follow-up) concludes. Results-disclosure timing is a separate, later milestone from either.

Multiplicity

The statistical problem created when a trial tests more than one comparison (e.g., two active dose arms vs. one placebo) without adjustment, which inflates the chance of a false-positive result somewhere in the set of tests.

REFERENCE

Appendix E Source Library

Every primary source and project file drawn on for this report, grouped by type.

Primary clinical publications

1. St. Clair EW, et al. Dazodalibep Phase 2 trial. *Nature Medicine*, 2024. DOI 10.1038/s41591-024-03009-3.
2. Bowman SJ, et al. Ianalumab (VAY736) Phase 2b dose-response trial. *The Lancet*, 2021. PMID 34861168.
3. Fisher BA, et al. Iscalimab (CFZ533) TWINSS Phase 2 trial. *The Lancet*, 2024. PMID 39096929.
4. TRACTISS primary trial report (rituximab). UCL Discovery repository.
5. Felten R, et al. ETAP trial (tocilizumab). PMID 33208345.
6. van Nimwegen JF, et al. ASAP-III trial (abatacept).
7. Novartis immunology portfolio update presentation (NEPTUNUS-1/2 conference data), slide 30.
8. ACR Convergence 2025 presentation, NEPTUNUS-1 and NEPTUNUS-2 results.

Registry sources

1. ClinicalTrials.gov exports: NCT06104124, NCT06245408, NCT04129164, NCT06747949 (dazodalibep program).
2. ClinicalTrials.gov exports: NCT05350072, NCT05349214, NCT05985915, NCT07621809, NCT02962895, NCT02149420, NCT05124925, NCT06293365, NCT02495129, NCT06411639 (ianalumab program).
3. ClinicalTrials.gov exports: NCT03905525, NCT02291029, NCT04541589 (iscalimab program).

Project working files (this project's own research and analysis)

1. README.md, PROJECT_PLAN.md: project scope and locked-in decisions.
2. 02_research/essdai_measurement_properties.md, phase2_baseline_characteristics.md, phase2_paper_explained.md, phase2_results.md, historical_trial_evidence.md, ianalumab_competitor_analysis.md, nct06245408_risk_note.md.
3. 01_data/NCT06104124.csv, NCT06245408.csv, competitor_trials_landscape.csv.
4. 03_analysis/evidence_table.md, scenario_model.md, final_thesis.md, additional_topics.md.
5. 00_reference/external_review.md: the independent review underlying Appendix C.
6. 04_summaries/plain_english_summary.md.

This library reflects sources as verified through the project's analysis cutoff of September 15, 2026. ClinicalTrials.gov listings are dynamic and should be re-verified before any investment or clinical decision relying on this report.